

Graphical Presentation of Statistical Data

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Graphs are visual representations of organized data.

Bar Graph

A **Bar Graph** is used to compare quantities among different categories.

For example, it can compare the number of students who prefer different sports.

Line Graph

A **Line Graph** is used to show changes or trends over time.

For example, it can show the number of books read during several months.

Pie Graph

A **Pie Graph** or circle graph shows how each category represents a part of the whole.

The entire circle represents the whole data set.

Stem-and-Leaf Plot

A **Stem-and-Leaf Plot** organizes numerical data while preserving the original values.

For example:

21, 24, 26, 28

can be organized using 2 as the stem and 1, 4, 6, 8 as the leaves.

Examples

Example 1

A survey of favorite sports produced the following data:

Basketball = 18, Volleyball = 12

Soccer = 8, Badminton = 6

Which graph is most suitable to compare these categories?

Bar Graph

Example 2

The number of books read by Ana during the first six months is shown:

Month	<i>Jan</i>	<i>Feb</i>	<i>Mar</i>	<i>Apr</i>	<i>May</i>	<i>Jun</i>
Books	2	3	5	6	4	7

Which graph best represents this information?

Line Graph

Example 3

A pie graph shows that $\frac{1}{2}$ of the class prefers milk, $\frac{1}{4}$ prefers juice, and the rest prefer water. What fraction of the class prefers water?

$$\begin{aligned} 1 - \frac{1}{2} - \frac{1}{4} \\ = \frac{1}{4} \end{aligned}$$

Therefore, $\frac{1}{4}$ prefers water.

Example 4

Choose the most appropriate graph for each situation.

Comparing sales of four stores

Showing rainfall over twelve months

Displaying the parts of a school's budget

Organizing students' quiz scores

Practice Exercises

Practice 1

A survey of favorite sports produced the following data:

Basketball = 18, Volleyball = 12

Soccer = 8, Badminton = 6

Which graph is most suitable to compare these categories?

Practice 2

The number of books read by Ana during the first six months is shown:

Jan = 2, Feb = 3, Mar = 5

Apr = 6, May = 4, Jun = 7

Which graph best represents this information?

Practice 3

A pie graph shows that $\frac{1}{2}$ of the class prefers milk, $\frac{1}{4}$ prefers juice, and the rest prefer water. What fraction of the class prefers water?

Practice 4

Choose the most appropriate graph for each situation.

Comparing sales of four stores

Showing rainfall over twelve months

Displaying the parts of a school's budget

Organizing students' quiz scores

Activities

Activity 1

A survey of favorite pets produced the following data:

Dogs = 20, Cats = 15

Birds = 7, Fish = 5

Which graph is most suitable to compare these categories?

Activity 2

The number of plants grown by Ben during the first six months is shown:

Jan = 4, Feb = 5, Mar = 6

Apr = 7, May = 8, Jun = 10

Which graph best represents this information?

Activity 3

A pie graph shows that $\frac{3}{5}$ of the class prefers reading, $\frac{1}{5}$ prefers drawing, and the rest prefer sports.

What fraction of the class prefers sports?

Activity 4

Choose the most appropriate graph for each situation.

Comparing the number of vehicles in different towns

Showing electricity consumption for twelve months

Displaying the distribution of household expenses

Organizing examination scores

Thank you!